

Kiaan Session 16

Tuesday, September 8, 2026 10:41 PM

Pattern 4:-

R1 1
R2 12
R3 123
R4 1234
R5 12345

Observation

1
2
3
4
5
6

```
for row in range(1,6):
```

num = 1

```
for col in range(row):
```

print(num,end)

num += 1

print()

Pattern 5:-

R1 1
R2 22
R3 333
R4 4444
R5 55555

Observation

```
for row in range(1,6):
```

num = row

```
for col in range(row):
```

print(num,end)

print()

Ascii Value:-

A → 65
B → 66
C → 67



Pattern 6:-

R1 A
R2 AB
R3 ABC

```
for row in range(1,6):
```

character = 'A'

```
for col in range(row):
```

R1 A
 R2 AB
 R3 ABC
 R4 ABCD
 R5 ABCDE

Character = 'A'

```

for col in range(Row):
    print(characters)
  
```

characters = chr(ord(characters)+1)

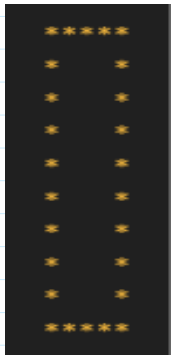
print()

Pattern 7:

A
 BB
 CCC
 DDDD
 EEEEE

→ Homework

Pattern 8: Hollow rectangle

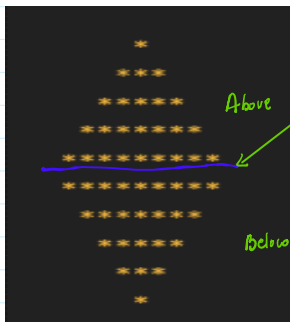


spaces

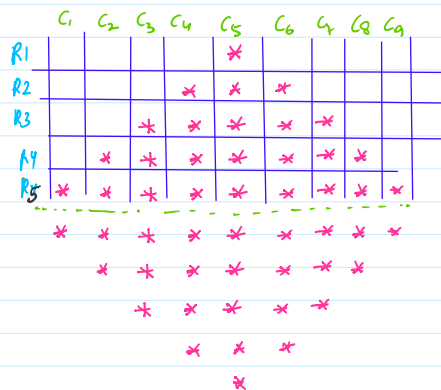
for (1,10):

* → R1 or R10
 * → C1 or C6
 otherwise → spaces

Pattern 9: Diamond Pattern



Above & Below → Mirror Images



spaces stars spaces

4 1 4
 3 3 3
 2 5 2
 1 7 1
 0 9 0

5-Row 2xRow-1 5-Row
 3-Row 2x2-1

Spaces Stars Spaces
 S-Row-num 2xRow-1 S-Row-num

→ 2xRowNum-1

* * *

5-Row-num | 2xRow-1 | 5-Row-num

R1

spaces	stars	spaces
5-1	2x1-1	5-1
4	2-1	4
	1	

→ 2xRowNum-1

R3

spaces	stars
5-3	2x3-1
2	5

// Above Part

for row in range(1:6):

excluded → 5 times

1. Spaces

```
for col in range(5-Row):
    print(" ", end="")
```

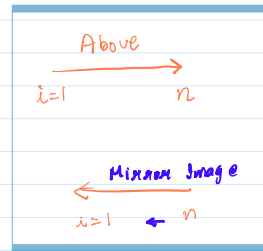
2. Stars

```
for col range(2xRow-1):
    print("*", end="")
```

3. Spaces

```
for col in range(5-Row):
    print(" ", end="")
```

Mirror Image → Change the direction of loop



Pattern 10 : Butterfly

Pattern 10 : Butterfly



Stars	Spaces	Stars
1	7	1
2	5	2
3	3	3
4	1	4
5	0	5
↓		
Row Number		
Total $2 \times (\text{Row} - i)$		
$2 \times (5 - 1)$		
$2 \times 4 = 8$		